# ITN 7.02 – Case Study - Part 4 (Modules 11 – 13)

Your supervisor has asked you to prepare a fourth part for the network simulation and presentation you have been working on. You will continuing with building the network you created for Part 3. You will add on to this file for Part 4. Open the Packet Tracer file **ITN – Part 4**. Be sure to do a **Save As** to rename the new file. A presentation will be used to demonstrate to the potential customer how your company would design and implement the proposed network. You have been asked to design this project in several segments that can ultimately be assembled into the full network topology.

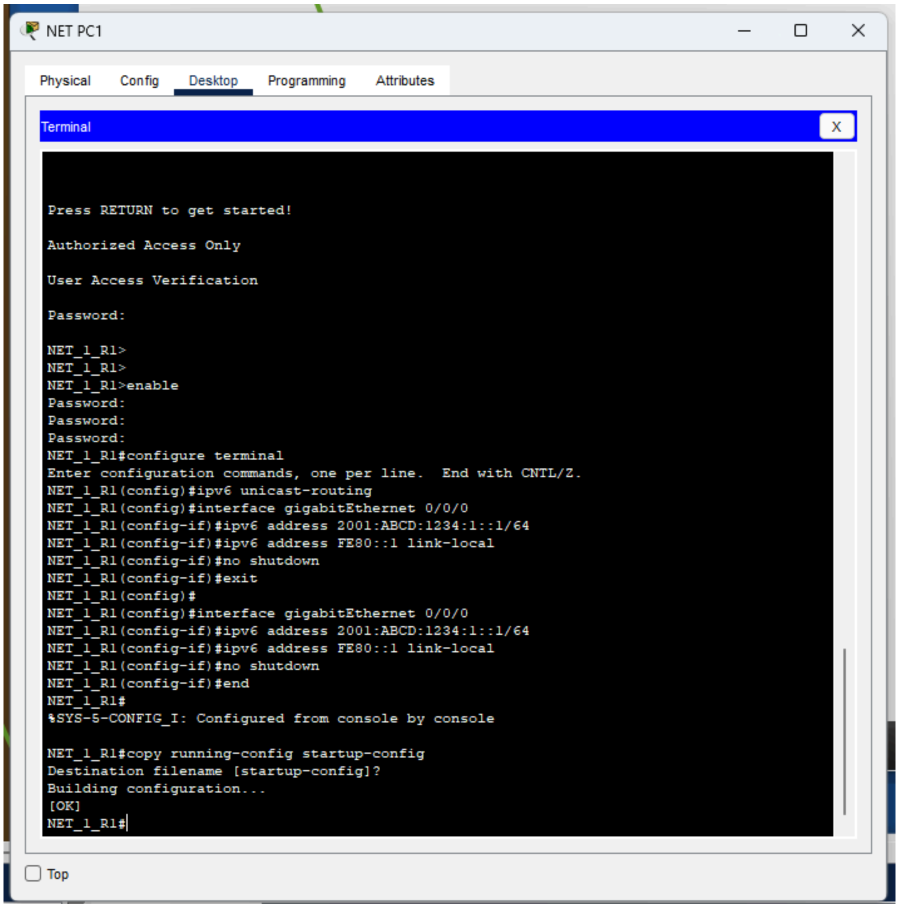
## Part 4 (Modules 11 – 13)

**Step 1 – Designing and documenting an IPv4 subnet scheme for a future growth network**

- You have been asked to plan an IPv4 address scheme for a future network.
- You have been given the IPv4 address range **192.168.1.0 /24** to address the network.
- Use this address space to create an address table bases on the following requirements:
  - LAN-A will need **60** host addresses.
  - LAN-B will need **25** host addresses.
  - LAN-C will need **20** host addresses.
  - LAN-D will need **12** host addresses.
  - LAN-E will need **12** host addresses.
  - The network will also need to address four point-to-point links.
- Include this table in your presentation and be prepared to explain how you develop it.

**Step 2 – Configuring IPv6 on the router**

- Access the router **NET_1_R1** via the **NET PC1** terminal program.
- Enable IPv6 routing
- Configure interface G0/0/0 with an IPv6 address
  - GUA – 2001:ABCD:1234:1::1 /64
  - Link local – FE80::1
- Configure interface G0/0/1 with an IPv6 address
  - GUA – 2001:ABCD:1234:2::1 /64
  - Link local – FE80::1




จากภาพจจะดเป็นขั้นตอนการทำงานโดย

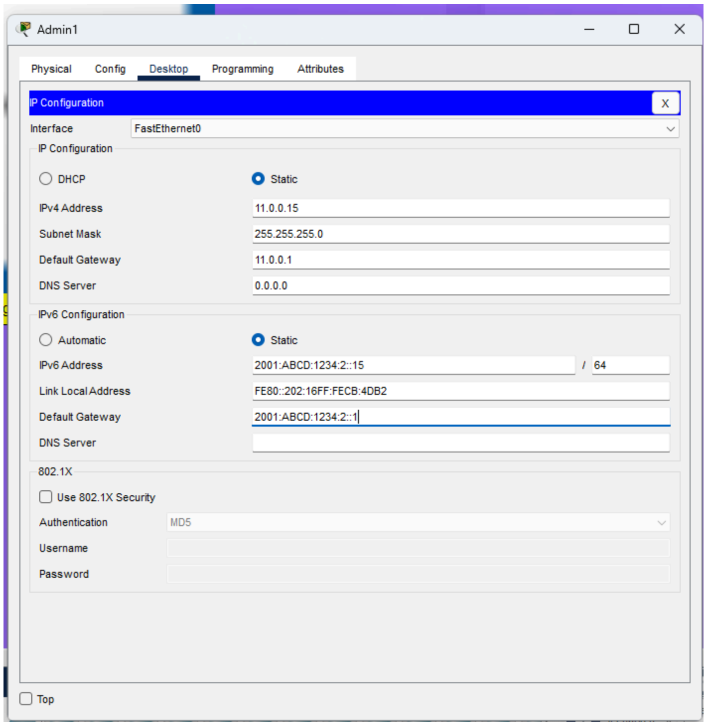
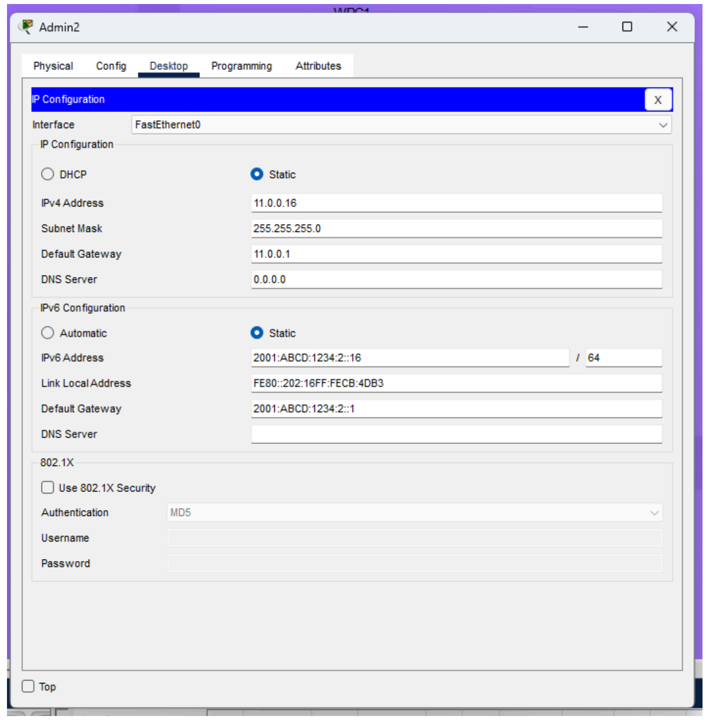
1. เข้าสู่ระบบ: เชื่อมต่อเข้าเราเตอร์ชื่อ NET_1_R1 ผ่านคอมพิวเตอร์ และใส่รหัสผ่านเพื่อเข้าโหมดตั้งค่า (configure terminal)
2. เปิดระบบ IPv6: ใช้คำสั่ง ipv6 unicast-routing เพื่อสั่งให้เราเตอร์รองรับระบบ IPv6
3. ตั้งค่าพอร์ตอินเทอร์เน็ต: เข้าไปที่พอร์ต gigabitEthernet 0/0/0 แล้วกำหนดหมายเลข IPv6 สองแบบคือ Global Address (2001:ABCD:1234:1::1/64) และ Link-Local Address (FE80::1)
4. เปิดใช้งานพอร์ต: ใช้คำสั่ง no shutdown เพื่อเปิดให้พอร์ตเริ่มทำงาน
5. บันทึกข้อมูล: ใช้คำสั่ง copy running-config startup-config เพื่อเซฟค่าทั้งหมดลงเราเตอร์ ป้องกันค่าหายเมื่อไฟดับ สังเกตสถานะเสร็จสิ้นที่คำว่า [OK]

**Step 3 –Configure PCs with an IPv6 address**

- PC1
  - GUA – 2001:ABCD:1234:1::15 /64
  - Link local – FE80::15
  - Use the address of the G0/0/0 interface as the default gateway
- PC2
  - GUA – 2001:ABCD:1234:1::16 /64
  - Link local – FE80::16
  - Use the address of the G0/0/0 interface as the default gateway

- PC3
  - GUA – 2001:ABCD:1234:1::17 /64
  - Link local – FE80::17
  - Use the address of the G0/0/0 interface as the default gateway
- PC4
  - GUA – 2001:ABCD:1234:1::18 /64
  - Link local – FE80::18
  - Use the address of the G0/0/0 interface as the default gateway
- PC5
  - GUA – 2001:ABCD:1234:1::19 /64
  - Link local – FE80::19
  - Use the address of the G0/0/0 interface as the default gateway
- Admin1
  - GUA – 2001:ABCD:1234:2::15 /64
  - Link local – FE80::15
  - Use the address of the G0/0/1 interface as the default gateway
- Admin2
  - GUA – 2001:ABCD:1234:2::16 /64
  - Link local – FE80::16
  - Use the address of the G0/0/1 interface as the default gateway

ภาพแสดงการIP Config ของ AdminและPC


Admin1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
11.0.0.15
Subnet Mask
255.255.255.0
Default Gateway
11.0.0.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
2001:ABCD:1234:2::15
/ 64
Link Local Address
FE80::202:16FF:FECB:4DB2
Default Gateway
2001:ABCD:1234:2::1
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top

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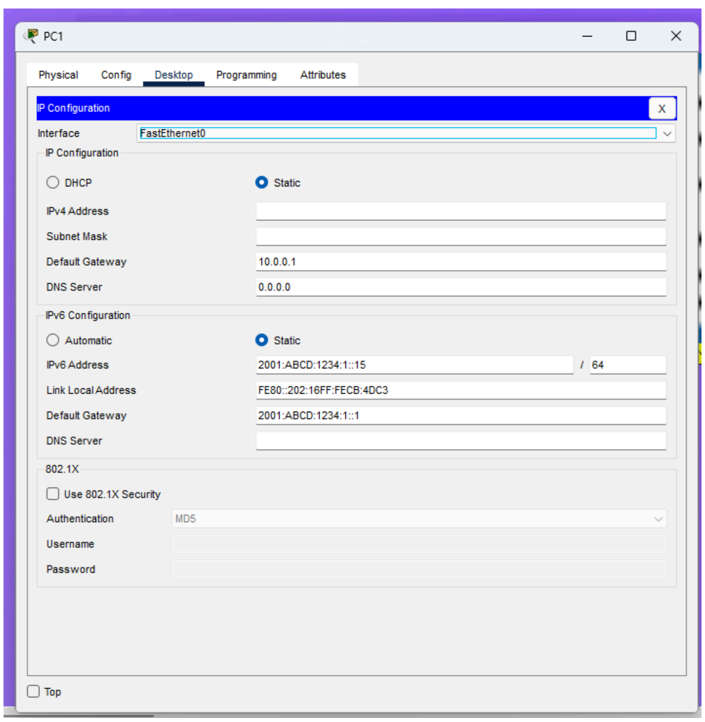
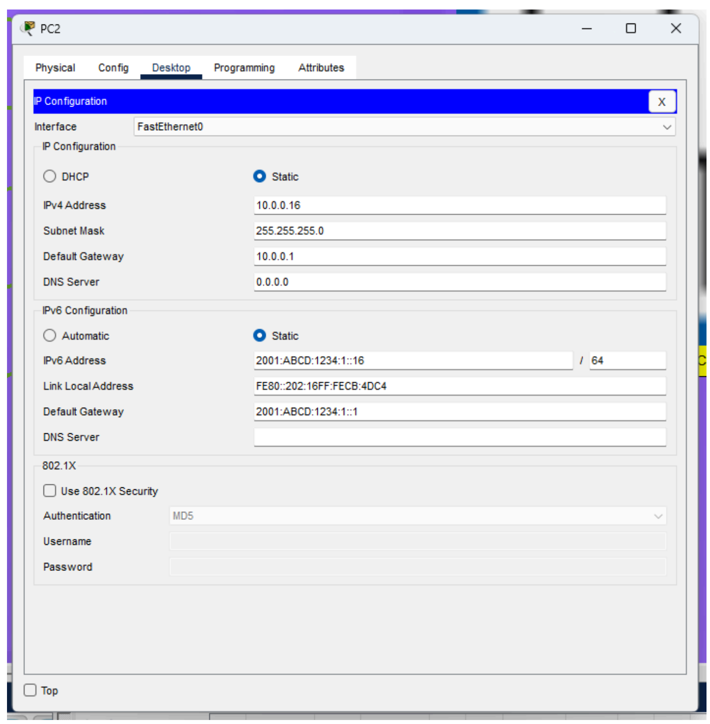
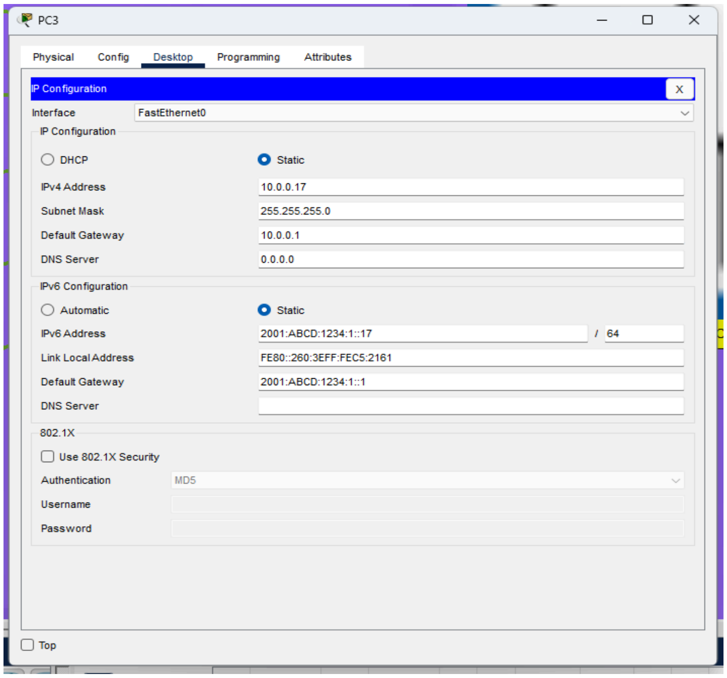
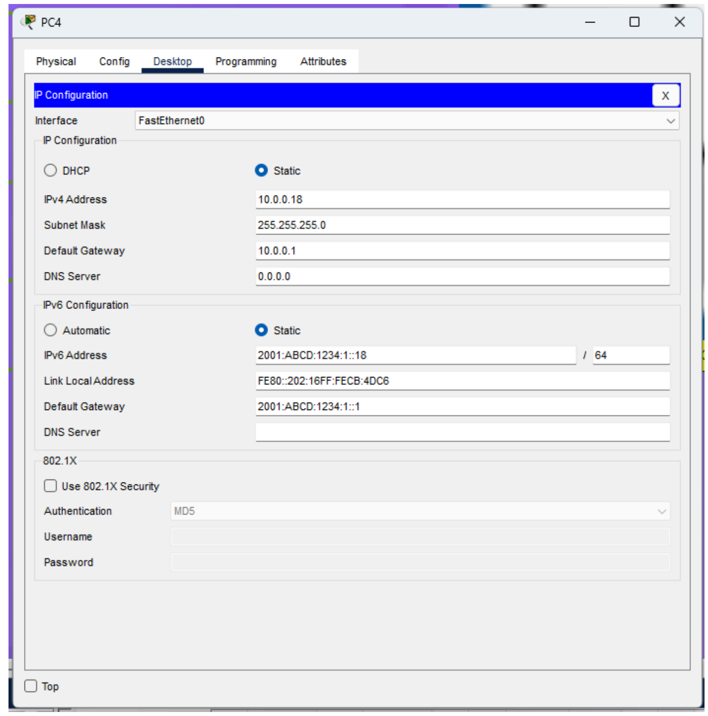
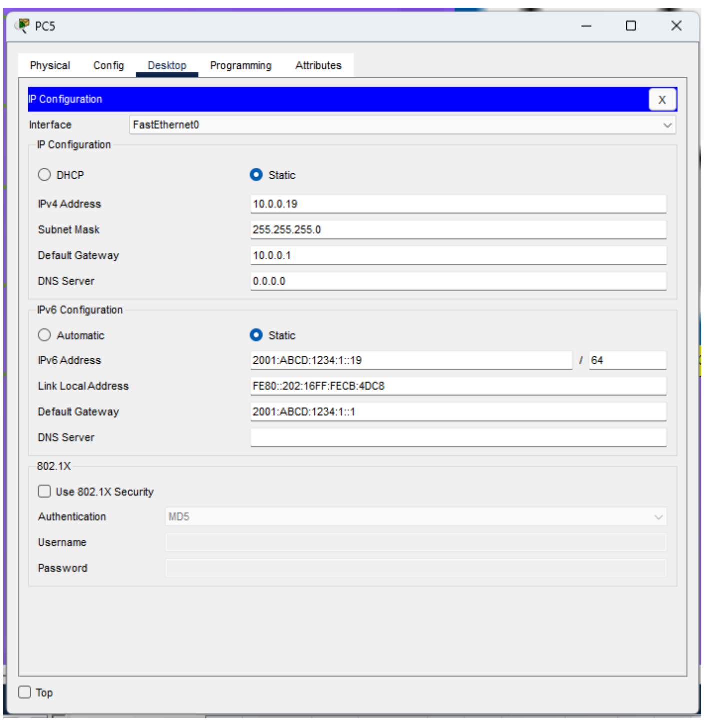
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
11.0.0.16
Subnet Mask
255.255.255.0
Default Gateway
11.0.0.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
2001:ABCD:1234:2::16
/ 64
Link Local Address
FE80::202:16FF:FECB:4DB3
Default Gateway
2001:ABCD:1234:2::1
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top

PC1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
Subnet Mask
Default Gateway
10.0.0.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
2001:ABCD:1234:1::15
/ 64
Link Local Address
FE80::202:16FF:FECB:4DC3
Default Gateway
2001:ABCD:1234:1::1
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top

PC2
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
10.0.0.16
Subnet Mask
255.255.255.0
Default Gateway
10.0.0.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
2001:ABCD:1234:1::16
/ 64
Link Local Address
FE80::202:16FF:FECB:4DC4
Default Gateway
2001:ABCD:1234:1::1
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top



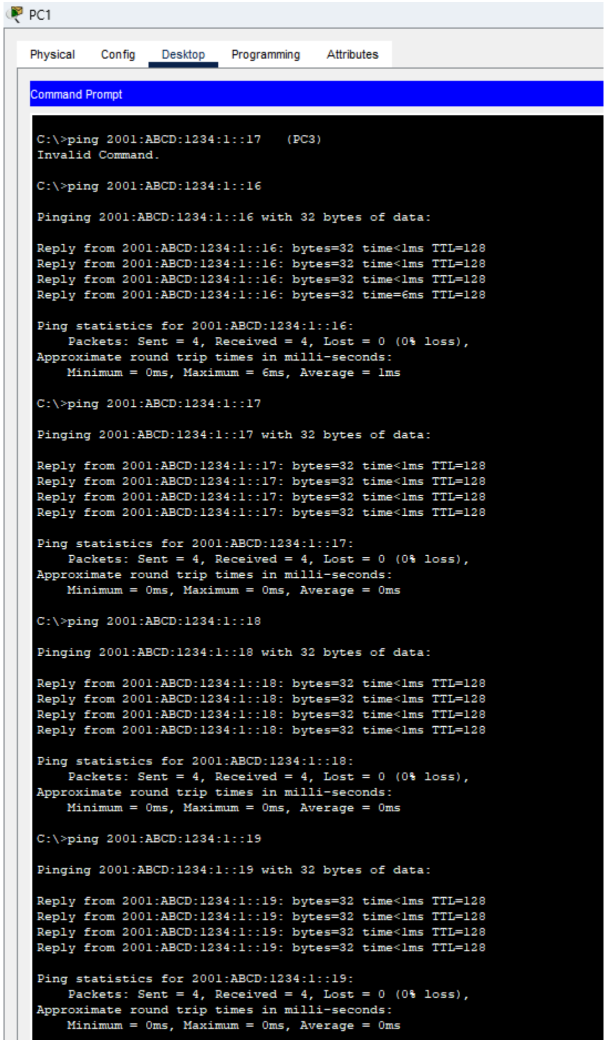



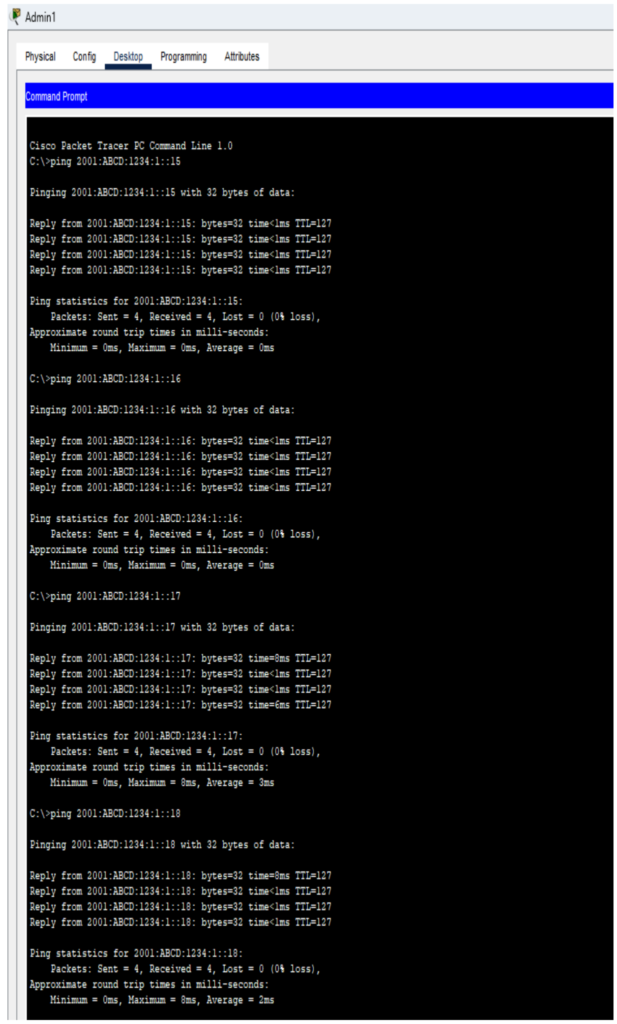




### Step 4 – Verify IPv6 connectivity

- Use PING to verify host-to-host connectivity.
- From PC1 test IPv6 connectivity to all other PCs with IPv6 addresses.
- From PC2 test IPv6 connectivity to all other PCs with IPv6 addresses.
- From PC3 test IPv6 connectivity to all other PCs with IPv6 addresses.
- From PC4 test IPv6 connectivity to all other PCs with IPv6 addresses.
- From PC5 test IPv6 connectivity to all other PCs with IPv6 addresses.

- From Admin1 test IPv6 connectivity to all other PCs with IPv6 addresses.
- From Admin2 test IPv6 connectivity to all other PCs with IPv6 addresses.

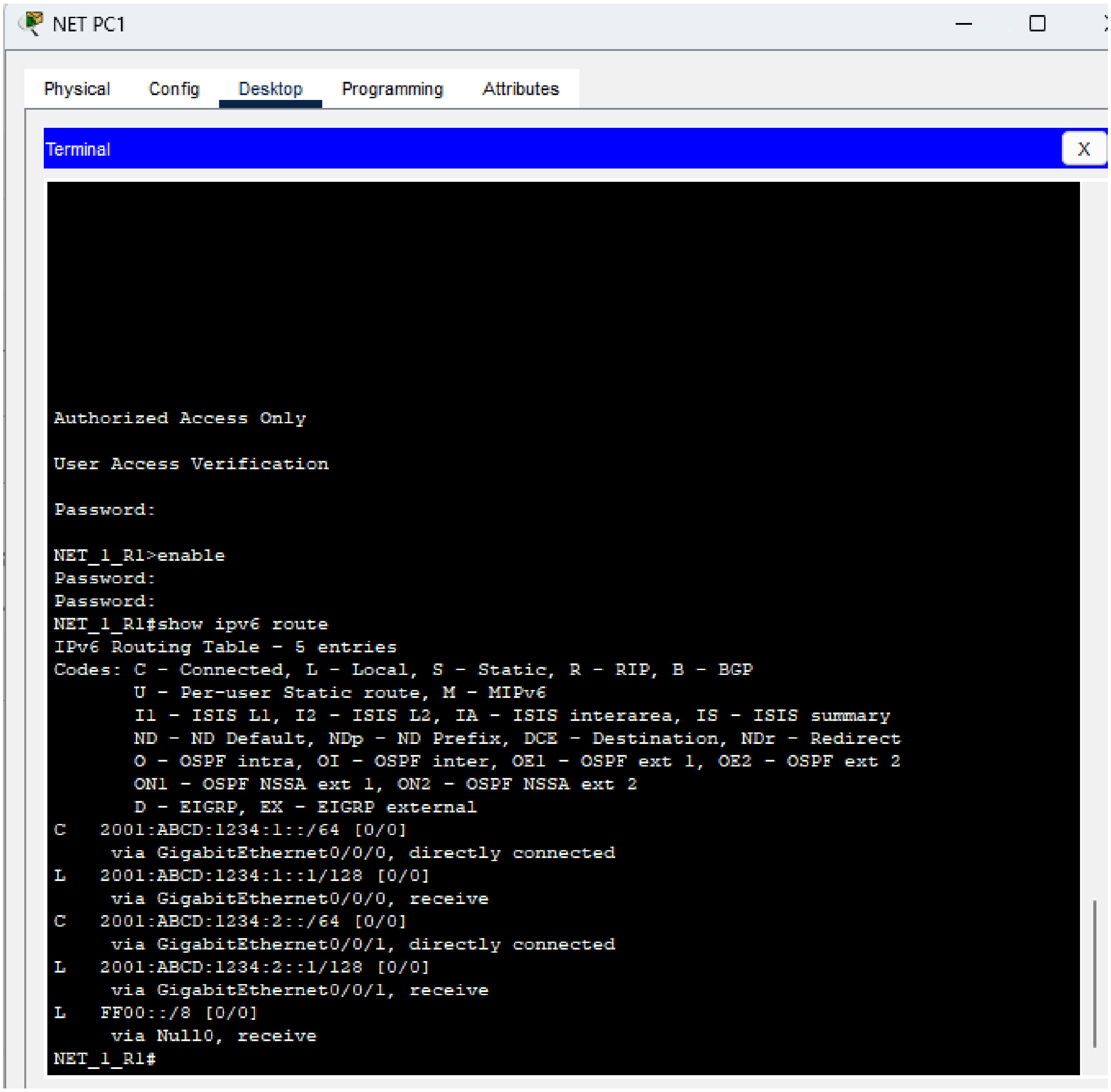



จากรูปคือตัวอย่างการ Ping เช็ค PC อื่นๆ

**Step 5 – Verify IPv6 routing**

- Verify the IPv6 routing table on **NET_1_R1**.




**Step 6 – Documentation**

- Add to the documentation you previously created of the physical network. Remember you will be adding to this document throughout the case study, so consider how you want it to be laid out as the potential customer will likely want to review the document.
- Document the current IPv6 routing table.
- Document the configuration of **NET_1_R1** (text copy).
- Document all IP addressing on the PCs.

**Step 7 – Additional information for presentation**

- Explain the role of the subnet mask when assigning IPv4 addresses.
- Explain how to determine what network a host belongs to.

- Explain the differences in
    - Unicast transmissions
    - Multicast transmissions
    - Broadcast transmissions
- Explain why IPv6 addressing is needed.
- Explain the difference between an IPv6 Global Unicast Address and an IPv6 Link-local Address.
- Describe the IPv6 Global Unicast Address structure and the role each part plays.
- Explain how host-to-host connectivity is tested and how the PING command works.
- Explain the difference in PING and TRACERT. When would you use one over the other?

**Step 8 – Presentation**

- Present your completed project in a presentation format to your workgroup and demonstrate full connectivity.